Risk Factors for Diabetes Among Adults Aged 40+

A Combined Logistic Regression and Random Forest Analysis Using NHANES Data

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Abstract

Diabetes disproportionately affects U.S. adults aged 40 years and older, yet the combined influence of socioeconomic, behavioral, and psychosocial determinants remains understudied. Using data from 2,873 adults in the 2009–2012 NHANES cycles, this study examined (1) how these factors predict diabetes risk and (2) whether variable importance rankings differ between logistic regression and a Random Forest model. After extensive data cleaning, transformation, and KNN imputation, we fitted a multivariable logistic regression model and trained a Random Forest classifier with cross-validation and downsampling to address class imbalance. Logistic regression identified age, race, low income, home rental, short sleep duration, sleep trouble, computer use, and depressive symptoms as significant predictors of diabetes. Random Forest variable importance rankings showed strong concordance with logistic regression—particularly for age, race, poverty, gender, and sedentary behaviors—while highlighting potential nonlinear effects for marital status, physical activity, and smoking. Together, findings underscore the multifactorial nature of diabetes risk and the value of integrating statistical and machine-learning approaches to identify both linear and complex determinants of disease among aging adults.

Introduction

Diabetes mellitus remains one of the most pressing chronic diseases affecting U.S. adults, with prevalence increasing substantially with age. According to the CDC's National Diabetes Statistics Report, 18.9% of adults aged 45-64 and 29.2% of adults aged 65 years and older have diabetes, reflecting an urgent need to understand the factors that place middle-aged and older adults at elevated risk (Centers for Disease Control and Prevention, 2024). Type 2 diabetes, which accounts for the vast majority of cases, develops through a multifactorial interplay of behavioral, psychosocial, metabolic, and socioeconomic determinants. As the population ages, identifying modifiable risk factors that influence diabetes onset among adults aged 40 and

older is critical for guiding effective clinical screening, public health planning, and targeted intervention strategies.

A substantial literature has examined the role of socioeconomic status, lifestyle factors, and psychosocial stressors in shaping diabetes risk. Socioeconomic status, for example, has consistently been recognized as one of the strongest predictors of diabetes. Analysis using national socioeconomic survey data demonstrated that gender, age, race, education, marital status, occupation, income, and debt levels were significantly associated with diabetes prevalence (Suwannaphant et al., 2017). Similarly, a 2023 study using NHANES data found a negative nonlinear association between income-to-poverty ratio and diabetes risk, independent of demographic and behavioral confounders, indicating that participants with lower socioeconomic status had a higher risk of diabetes (Liu et al., 2023). Behavioral factors, including physical inactivity, sleep quantity or quality, alcohol consumption patterns, and smoking, also play a well-established role in diabetes development (Ismail et al., 2021). Sleep duration, in particular, has been shown to exhibit a U-shaped relationship with diabetes risk: both short sleep (<6 hours) and long sleep (>9 hours) are associated with elevated odds compared to adequate sleep (Cappuccio et al., 2010; Shan et al., 2015). Similarly, while moderate alcohol consumption has been hypothesized to confer metabolic benefits in some populations, observational studies face challenges in disentangling causal effects from confounding by socioeconomic status and lifestyle patterns. Recent research has also highlighted the role of psychosocial determinants, demonstrating that symptoms of depression and anxiety contribute meaningfully to diabetes risk, potentially through both behavioral pathways (e.g., poor adherence to healthy lifestyles) and physiological mechanisms (e.g., chronic stress-induced metabolic dysregulation) (Naicker et al., 2018). Despite extensive evidence, relatively few studies simultaneously examine socioeconomic, behavioral, and psychosocial factors within a single representative dataset for U.S. adults aged 40 and older.

Furthermore, most prior studies have relied exclusively on traditional regression-based approaches, which, while interpretable, may overlook complex interactions and nonlinear relationships among risk factors. Machine learning methods, such as random forests, offer complementary advantages by capturing intricate patterns without assuming linearity or requiring explicit specification of interaction terms. Comparing the variable importance rankings from machine learning models with the statistically significant predictors identified through logistic regression can provide deeper insights into which factors most reliably predict diabetes risk across different analytical frameworks.

The present study addresses these gaps by leveraging data from the National Health and Nutrition Examination Survey (NHANES) to investigate two primary research questions: (1) Among U.S. adults aged 40 years and older, how do socioeconomic, behavioral, and psychosocial factors influence the risk of diabetes? (2) How do the variable importance rankings from a random forest model compare with the statistically significant predictors from logistic regression? By integrating classical statistical inference with modern machine learning, this study aims to provide a comprehensive, multidimensional assessment of diabetes risk factors, with implications for targeted prevention and intervention strategies in an aging population.

Methods

Data Source and Study Population

The National Health and Nutrition Examination Survey (NHANES) offers an ideal platform for addressing this research question. NHANES integrates clinically validated biomarker measurements, standardized physical examinations, and detailed demographic, behavioral, and psychosocial survey data, enabling a comprehensive, multidimensional analysis of diabetes risk factors. In this project, we utilized the NHANES dataset available through the NHANES R package, which provides a streamlined version of 10,000 observations resampled from NHANESraw to mitigate oversampling effects inherent in the original survey design. The data include variables from the 2009-2010 and 2011-2012 survey cycles, comprising 76 variables across demographic, socioeconomic, behavioral, physiological, and psychosocial domains (Supplementary Table 1).

Because our primary objective is to model and compare predictors rather than estimate nationally weighted prevalence, the NHANES package is more suitable than NHANESraw; it provides a structure conducive to both classical statistical modeling and modern machine-learning approaches while avoiding the survey-weight complexity required for population-level inference. We restricted our analytic sample to adults aged 40 years and older, as this age group bears the highest diabetes burden and represents the target population for most prevention and screening programs. After removing duplicate observations (identified by participant ID) and excluding pregnant women (whose metabolic profiles differ substantially from the non-pregnant population), our final analytic dataset consisted of 2,873 participants.

Data Cleaning and Preprocessing

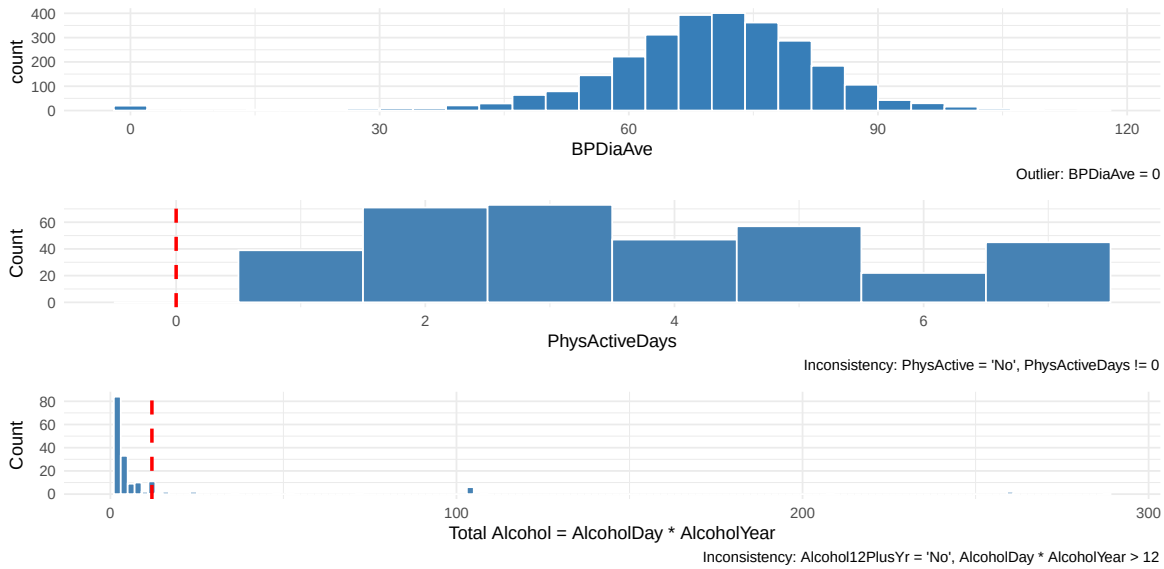


Figure 1: Data Quality: Detecting Outliers and Inconsistencies

Figure 1 revealed several data quality issues requiring correction prior to modeling. First, diastolic blood pressure (BPDiaAve) contained biologically implausible values of zero, which we recoded as missing (NA) to avoid spurious associations. We also identified logical inconsistencies in physical activity reporting: some participants indicated no moderate or vigorous-intensity physical activity (PhysActive = "No") but reported positive values for PhysActiveDays. We corrected this by setting PhysActiveDays to zero for all participants who reported no physical activity. Additionally, alcohol consumption variables showed inconsistencies: participants who consumed less than 12 drinks of any type of alcoholic beverage in any one year (Alcohol12PlusYr = "No") sometimes had AlcoholDay × AlcoholYear (Total Alcohol Consumption) values exceeding 12 drinks. We recoded these implausible values as missing.

In our dataset, missing data varied substantially across variables (Supplementary Table 2). PhysActiveDays and SmokeNow each had over 50% missingness, AlcoholDay and AlcoholYear exceeded 20%, and Race3, TVHrsDay, and CompHrsDay exhibited complete missingness for the 2009-2010 survey cycle due to data collection artifacts. To retain all 2,873 participants and maximize statistical power, we employed k-nearest neighbors (KNN) imputation (k = 5) using the VIM package. KNN imputes missing values by identifying the k most similar observations based on Euclidean distance and using their mean (numerical) or median/mode (categorical) values. Unlike single-value methods such as mean or median imputation, KNN preserves correlations between variables.

However, KNN imputation has important limitations. It assumes missingness is at random (MAR) meaning that missingness must depend only on observed variables; violations of this assumption may introduce bias. In addition, KNN is sensitive to the choice of k: small values may overfit noise, while large values may oversmooth important structure. We selected k = 5 as a balanced choice. Despite these limitations, KNN was preferable to complete-case analysis (which would discard >40% of the sample) or mean/median imputation (which attenuates relationships).

Variable Selection and Transformation

Our variable selection was guided by theoretical relevance to diabetes, prior literature, and avoidance of post-treatment or mediator bias. We excluded variables not applicable to adults aged 40+, including child anthropometrics (Length, HeadCirc, BMICatUnder20yrs, TVHrsDayChild, CompHrsDayChild) and pregnancy-related measures (nPregnancies, nBabies, Age1stBaby). Variables with limited relevance to diabetes, such as testosterone, urinary flow/volume measures, marijuana-use indicators, and sexual behavior variables, were also removed. To prevent mediator adjustment, we excluded physiological measures that likely lie on the causal pathway (BMI, Pulse, BPSysAve, BPDiaAve, DirectChol, TotChol, repeated BP readings). Post-treatment health indicators (HealthGen, DaysPhysHlthBad, DaysMentHlthBad) were omitted to avoid collider bias. Redundant variables including Race1, AgeDecade, AgeMonths, HHIncome, HHIncomeMid, HomeRooms, Weight, Height, BMI_WHO, DiabetesAge, PhysActiveDays, SmokeAge, and Smoke100n were removed due to overlap with retained measures. This process yielded a final set of socioeconomic, behavioral, and psychosocial predictors with strong theoretical justification. Figure 2 summarizes the distribution of selected variables.

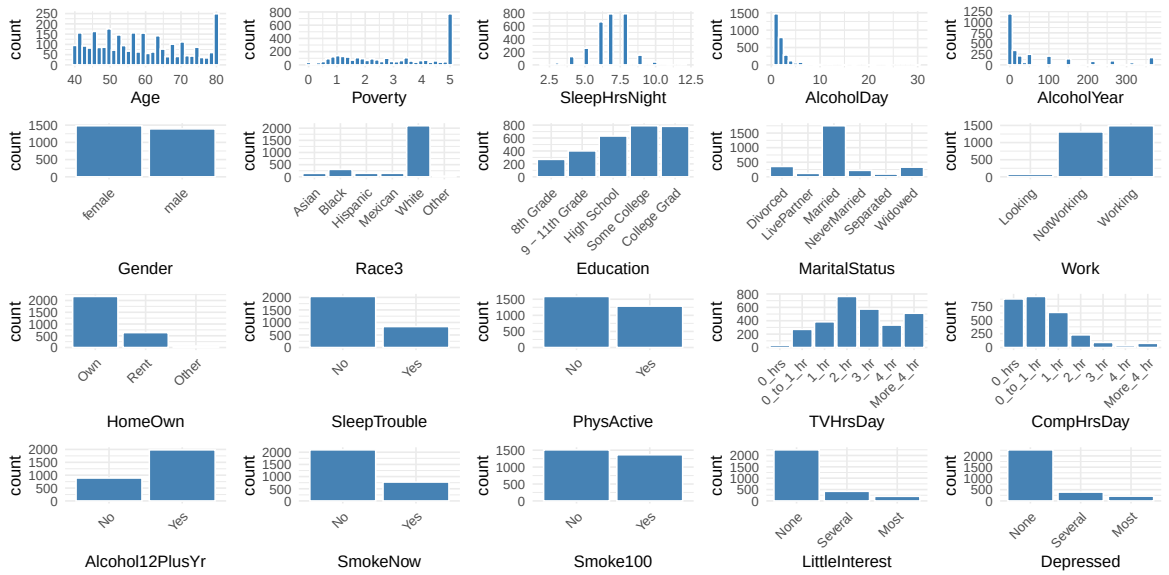


Figure 2: Distribution of Numeric and Categorical Variables)

Retained variables were then harmonized and recoded. Following prior evidence of nonlinear income–diabetes associations (Liu et al., 2023), Poverty (IPR) was categorized into Low (1.0), Middle (1.0–4.0), and High (4.0) income groups. Education was collapsed into High School, Some College, and College+, and MaritalStatus into Never Married, Married, and Not Married (Divorced, Live Partner, Separated, Widowed). SleepHrsNight, which demonstrates a U-shaped association with diabetes (Cappuccio et al., 2010; Shan et al., 2015), was categorized into Short (<7 hours), Adequate (7–9 hours), and Long (>9 hours). Smoking status was derived from SmokeNow and Smoke100 as Never, Former, and Current. Alcohol use was quantified as annual drinks (AlcoholDay × AlcoholYear) and categorized into Non-drinker, Light–Moderate (<730 drinks/year), and Heavy (730 drinks/year). Ordinal sedentary behavior variables (TVHrsDay, CompHrsDay) were converted to numeric midpoints to approximate daily hours. Psychosocial variables (LittleInterest, Depressed) were dichotomized to reflect presence/absence of depressive symptoms.

Exploratory Data Analysis

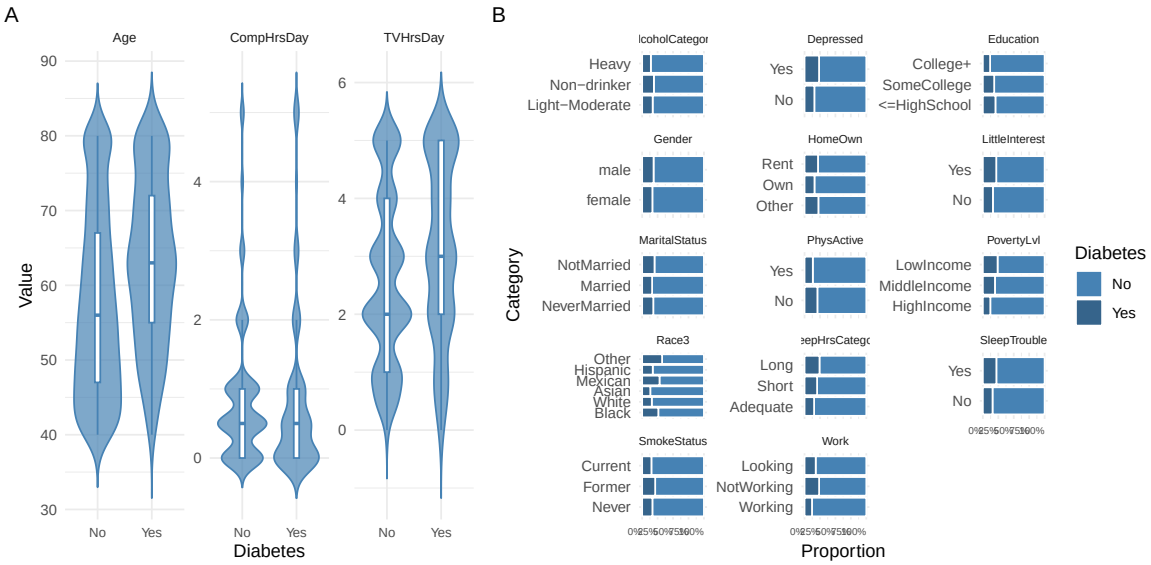


Figure 3: Numeric (Left) and Categorical Variables (Right) by Diabetes Risk

After variable transformation, we conducted exploratory data analysis in Figure 3 to assess distributions and multicollinearity. Violin-boxplot overlays for continuous variables showed that participants with diabetes were significantly older, while sedentary behavior variables (TVHrsDay, CompHrsDay) exhibited broad overlap between groups. Among categorical variables, diabetes prevalence was higher among those with lower education, lower income, sleep trouble, and physical inactivity. Former smokers showed higher diabetes prevalence than never smokers, consistent with reverse causation. Assessment of multicollinearity indicated no problematic correlations among retained variables (all pairwise correlations $r < 0.7$; Supplementary Figure 1), supporting their suitability for subsequent modeling.

Statistical Analysis

We employed two complementary analytical approaches to address our research questions and evaluate the robustness of identified predictors.

Logistic Regression

We fitted a multivariable logistic regression model with diabetes status (yes/no) as the binary outcome and all retained socioeconomic, behavioral, and psychosocial variables as predictors.

The model was specified as:

$$\log\left(\frac{P(\text{Diabetes} = \text{Yes})}{1 - P(\text{Diabetes} = \text{Yes})}\right) = \beta_0 + \beta_1 \text{Gender}_{\text{male}} + \beta_2 \text{Age} + \beta_3 \text{Race}_{\text{White}} + \beta_4 \text{Race}_{\text{Asian}} + \beta_5 \text{Race}_{\text{Mexican}}$$
$$+ \beta_6 \text{Race}_{\text{Hispanic}} + \beta_7 \text{Race}_{\text{Other}} + \beta_8 \text{Education}_{\text{SomeCollege}} + \beta_9 \text{Education}_{\text{College+}} + \beta_{10} \text{MaritalStatus}_{\text{Married}}$$
$$+ \beta_{11} \text{MaritalStatus}_{\text{NotMarried}} + \beta_{12} \text{Poverty}_{\text{MiddleIncome}} + \beta_{13} \text{Poverty}_{\text{LowIncome}} + \beta_{14} \text{Work}_{\text{NotWorking}}$$
$$+ \beta_{15} \text{Work}_{\text{Looking}} + \beta_{16} \text{HomeOwn}_{\text{Rent}} + \beta_{17} \text{HomeOwn}_{\text{Other}} + \beta_{18} \text{Sleep}_{\text{Short}} + \beta_{19} \text{Sleep}_{\text{Long}}$$
$$+ \beta_{20} \text{SleepTrouble}_{\text{Yes}} + \beta_{21} \text{PhysActive}_{\text{Yes}} + \beta_{22} \text{TVHrsDay} + \beta_{23} \text{CompHrsDay} + \beta_{24} \text{Smoke}_{\text{Former}}$$
$$+ \beta_{25} \text{Smoke}_{\text{Current}} + \beta_{26} \text{Alcohol}_{\text{NonDrinker}} + \beta_{27} \text{Alcohol}_{\text{Heavy}} + \beta_{28} \text{LittleInterest}_{\text{Yes}} + \beta_{29} \text{Depressed}_{\text{Yes}}$$

where reference categories for categorical variables: Female (gender), Black (race/ethnicity), High School (education), Never Married (marital status), High Income (poverty level), Working (employment), Own (home ownership), Adequate 7–9 hours (sleep duration), No (sleep trouble), Inactive (physical activity), Never Smoker (smoking status),Light-Moderate Drinker (alcohol consumption), No (little interest and depressed mood). Age, TVHrsDay, and CompHrsDay are continuous variables; odds ratios represent the change in diabetes odds per one-year increase in age and per one-hour increase in screen time.

Logistic regression was selected for its ability to provide interpretable odds ratios (ORs) with corresponding confidence intervals, enabling identification of statistically significant predictors and quantification of their associations with diabetes risk. Model diagnostics included variance inflation factors (VIFs) to assess multicollinearity and standard diagnostic plots to evaluate overall model adequacy.

Random Forest

To complement the parametric logistic regression model, we trained a Random Forest classifier using the same set of predictors. Random forests are ensemble learning methods that aggregate predictions from multiple decision trees, each trained on a bootstrap sample of the data. Unlike logistic regression, random forests do not assume linearity, making them well-suited for capturing complex, nonlinear relationships. The model was trained using five-fold cross-validation via the caret framework. Hyperparameter tuning was performed over candidate values of the `mtry` parameter, representing the number of variables randomly sampled at each split. Model selection was based on area under the ROC curve (AUC). Class imbalance was addressed by downsampling the majority class during training, ensuring balanced class frequencies within cross-validation folds. Model evaluation was performed on an independent hold-out test set. Performance metrics included ROC–AUC, Precision–Recall AUC (PR–AUC), sensitivity, specificity, and balanced accuracy. ROC–AUC measures overall discrimination, whereas PR–AUC provides a more sensitive assessment under class imbalance by emphasizing the tradeoff between false positives and recall of diabetic cases.

Comparison

We compared the top 15 predictors from Random Forest (ranked by permutation-based importance) with the top 15 predictors from logistic regression (ranked by |z-value|). Concordance between methods was assessed by examining overlap in the top-ranked variables and noting areas of agreement and divergence. High agreement between parametric (logistic regression) and non-parametric (Random Forest) approaches would strengthen confidence in the robustness of identified risk factors, while divergence may indicate nonlinear relationships or interactions that logistic regression does not capture.

[**need check and revise**] [To complement the logistic regression analysis, a Random Forest classifier was implemented to evaluate predictive performance and identify influential variables without imposing parametric assumptions. The model was trained using five-fold cross-validation via the caret framework. Hyperparameter tuning was performed over candidate values of the `mtry` parameter, representing the number of variables randomly sampled at each split. Model selection was based on area under the ROC curve (AUC). Class imbalance was addressed by downsampling the majority class during training, ensuring balanced class frequencies within cross-validation folds.

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Feature importance was examined using two methods. First, Gini importance quantified how frequently and effectively variables reduced node impurity across trees. Second, permutation-based importance was computed by randomly shuffling each variable and measuring the resulting decrease in ROC–AUC. This approach directly reflects changes in predictive performance and allows for meaningful ranking across heterogeneous features.

Random Forest was selected not as a replacement for logistic regression, but as a complementary tool to detect nonlinear relationships and interactions that parametric models may overlook. Agreement between regression coefficients and variable importance enhances confidence in model findings, while divergence may indicate higher-order relationships or multicollinearity effects.]

Results

Our final analytic sample comprised 2,873 adults aged 40 years and older, of whom 502 (17.5%) had diabetes and 2,371 (82.5%) did not.

Logistic Regression Results

Table 1 presents estimated coefficients, odds ratios (ORs) and 95% confidence intervals (CIs) for our multivariable logistic regression model. Model diagnostics confirmed the validity of results: variance inflation factors (VIF) were all below 5, indicating minimal multicollinearity (Supplementary Table S3), and standard diagnostic plots showed no major violations of model assumptions (Supplementary Figure S2).

Among demographic confounders, age exhibited the strongest association with diabetes risk, with each additional year associated with a 4% increase in the odds of diabetes (OR = 1.04, 95% CI: 1.03–1.05; $p < 0.001$). Significant racial and ethnic disparities were observed. Compared with Black participants, White (OR = 0.45, 95% CI: 0.33–0.62; $p < 0.001$), Asian (OR = 0.48, 95% CI: 0.26–0.84; p = 0.013), and Hispanic participants (OR = 0.55, 95% CI: 0.31–0.93; p = 0.028) had substantially lower odds of diabetes. Male participants had 33% higher odds of diabetes compared with females (OR = 1.33, 95% CI: 1.07–1.66; p = 0.012).

Among socioeconomic factors, income-to-poverty ratio demonstrated a graded association with diabetes risk. Participants in the low-income group had 49% higher odds of diabetes compared with those in the high-income group (OR = 1.49, 95% CI: 1.03–2.15; p = 0.034), while those in the middle-income group showed a non-significant trend toward higher odds (OR = 1.25, 95% CI: 0.97–1.63; p = 0.087). Renting a home, compared with homeownership, was associated with higher odds of diabetes (OR = 1.41, 95% CI: 1.08–1.83; p = 0.011). Education level, marital status, and employment status were not significantly associated with diabetes after adjustment.

Sleep-related variables emerged as important behavioral correlates of diabetes. Short sleep duration (<7 hours) was associated with higher odds of diabetes (OR = 1.26, 95% CI: 1.01–1.56; p = 0.037), and self-reported sleep trouble was also positively associated with diabetes risk (OR = 1.36, 95% CI: 1.08–1.71; p = 0.009). Among sedentary behaviors, computer usage hours exhibited a dose–response relationship, with each additional hour associated with a 13% increase in the odds of diabetes (OR = 1.13, 95% CI: 1.03–1.24; p = 0.010). Television viewing time showed a borderline association (OR = 1.07 per additional hour, 95% CI: 1.00–1.16; p = 0.061). Physical activity was associated with lower odds of diabetes, although this association did not reach statistical significance (OR = 0.80, 95% CI: 0.64–1.01; p = 0.057).

Regarding health behaviors, current smoking was associated with lower odds of diabetes compared with never smoking (OR = 0.74, 95% CI: 0.55–0.99; p = 0.046), whereas former smoking showed no significant association (OR = 1.07, 95% CI: 0.83–1.36; p = 0.607). Alcohol consumption categories were not significantly associated with diabetes risk.

Among psychosocial factors, self-reported depressive symptoms were associated with higher odds of diabetes (OR = 1.43, 95% CI: 1.07–1.90; p = 0.016), whereas little interest in activities was not significantly associated with diabetes (OR = 1.08, 95% CI: 0.81–1.43; p = 0.585).

Table 1: Logistic Regression Results for Diabetes Risk Factors

Variable	Estimate	z value	p-value	OR (95% CI)	
(Intercept)	-4.303	-9.756	<0.001	0.01 (0.01–0.03)	***
Gendermale	0.284	2.516	0.012	1.33 (1.07–1.66)	*
Age	0.040	6.827	<0.001	1.04 (1.03–1.05)	***
Race3White	-0.793	-4.949	<0.001	0.45 (0.33–0.62)	***
Race3Asian	-0.733	-2.473	0.013	0.48 (0.26–0.84)	*
Race3Mexican	0.260	1.063	0.288	1.30 (0.80–2.09)	
Race3Hispanic	-0.605	-2.203	0.028	0.55 (0.31–0.93)	*
Race3Other	0.250	0.577	0.564	1.28 (0.53–2.93)	
Education Some College	0.125	0.987	0.323	1.13 (0.88–1.45)	
Education College+	-0.079	-0.506	0.613	0.92 (0.68–1.25)	
Marital Status Married	0.023	0.112	0.911	1.02 (0.69–1.54)	
Marital Status Not Married	-0.105	-0.499	0.618	0.90 (0.60–1.37)	
Poverty Lvl Middle Income	0.226	1.709	0.087	1.25 (0.97–1.63)	.
Poverty Lvl Low Income	0.399	2.124	0.034	1.49 (1.03–2.15)	*
Work Not Working	0.215	1.611	0.107	1.24 (0.95–1.61)	
Work Looking	0.128	0.378	0.706	1.14 (0.56–2.14)	
Home Own Rent	0.341	2.537	0.011	1.41 (1.08–1.83)	*
Home Own Other	0.286	0.894	0.372	1.33 (0.69–2.44)	
Sleep Hrs Category Short	0.231	2.082	0.037	1.26 (1.01–1.56)	*
Sleep Hrs Category Long	0.129	0.404	0.686	1.14 (0.59–2.08)	
Sleep Trouble Yes	0.307	2.605	0.009	1.36 (1.08–1.71)	**
Phys Active Yes	-0.220	-1.905	0.057	0.80 (0.64–1.01)	.
TVHrs Day	0.072	1.873	0.061	1.07 (1.00–1.16)	.
Comp Hrs Day	0.121	2.584	0.010	1.13 (1.03–1.24)	**
Smoke Status Former	0.065	0.514	0.607	1.07 (0.83–1.36)	
Smoke Status Current	-0.298	-1.999	0.046	0.74 (0.55–0.99)	*
Alcohol Category Non-drinker	-0.032	-0.257	0.797	0.97 (0.76–1.23)	
Alcohol Category Heavy	-0.106	-0.414	0.679	0.90 (0.53–1.46)	
Little Interest Yes	0.079	0.546	0.585	1.08 (0.81–1.43)	
Depressed Yes	0.356	2.410	0.016	1.43 (1.07–1.90)	*

Random Forest Results

[need revise]

After correcting for class imbalance, the Random Forest classifier showed good predictive performance on the test dataset, achieving an ROC–AUC of 0.746. This indicates that the model had strong ability to distinguish between individuals with and without diabetes.

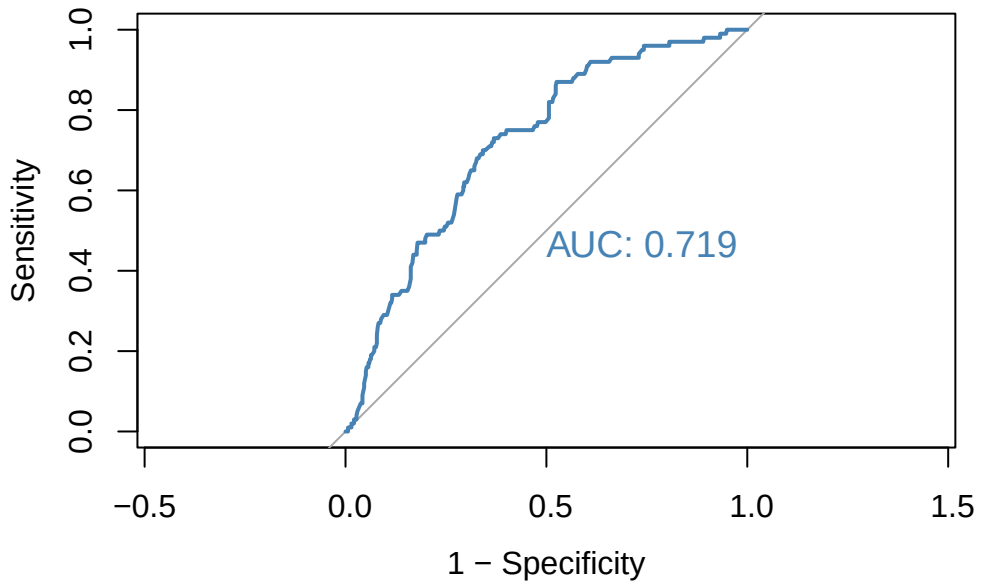


Figure 4: ROC Curve - Balanced Random Forest

Figure X. ROC curve for the balanced Random Forest classifier. The model demonstrates good discriminative ability, with an AUC of 0.746.

The Precision–Recall curve further illustrates classification performance in the presence of class imbalance.

```
plot(pr, main = sprintf("Precision-Recall Curve (AUC = %.3f)", pr$auc.integral))
```

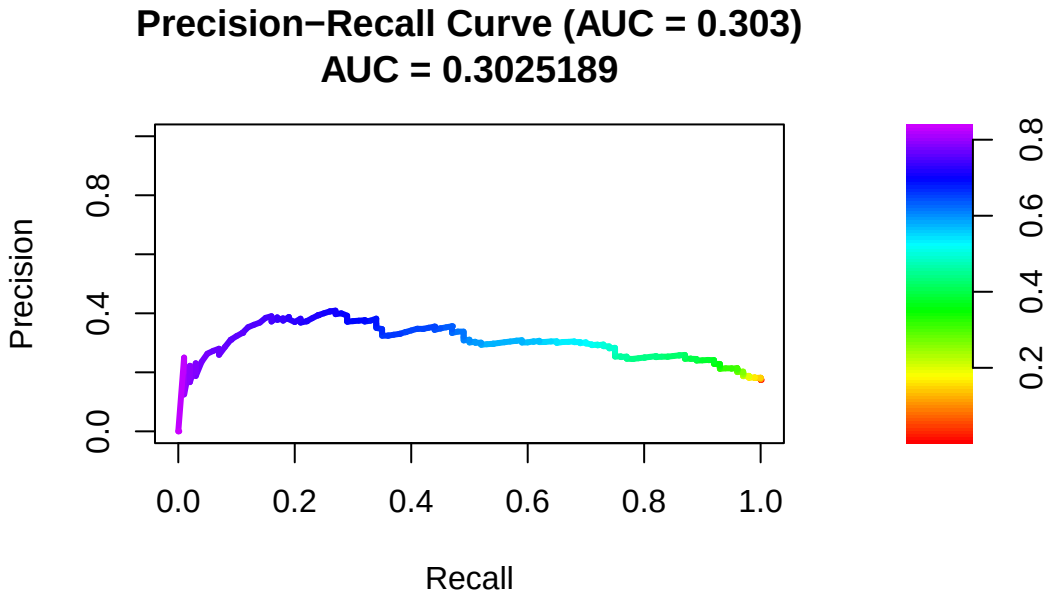


Figure X. Precision–Recall curve for diabetes classification. Performance reflects class imbalance, with declining precision as recall increases.

Test-set accuracy was 0.70, with a balanced accuracy of 0.71, indicating stable classification across outcome groups. Sensitivity reached 0.72, meaning the model correctly identified approximately 72% of individuals with diabetes. Specificity was 0.69, showing comparably strong detection of non-diabetic individuals.

The Precision–Recall AUC was 0.35. While lower than the ROC–AUC, this result is expected in imbalanced classification settings. With diabetes affecting only about 17% of the sample, precision naturally declines when recall increases. In this context, the observed PR-AUC reflects meaningful case-finding ability rather than poor model performance.

Although overall accuracy declined relative to the no-information rate, this was a direct consequence of balancing class weights. Sensitivity improved substantially relative to the unweighted model, reflecting an intentional shift toward identifying clinically relevant cases rather than maximizing raw accuracy.

Overall, the balanced Random Forest model achieved improved case detection while maintaining stable and interpretable performance.

Classification Performance

```
fourfoldplot(cm$table,
color = c("steelblue", "lightcoral"),
conf.level = 0,
margin = 2)
```

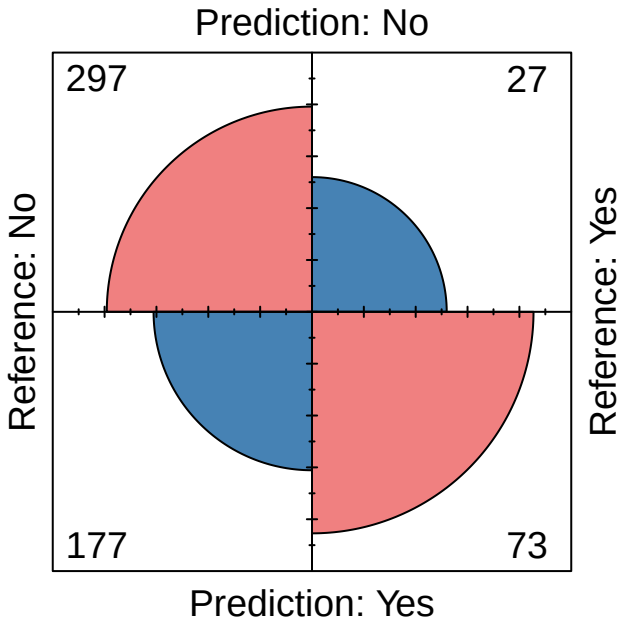


Figure 5: Confusion Matrix - Balanced Random Forest

The confusion matrix illustrates model behavior across both outcomes. While false positives increased as a result of rebalancing, false negatives decreased substantially. This tradeoff strengthens usefulness in clinical settings where missing diabetes cases is more costly than generating additional screenings.

Variable Importance from Random Forest

Two complementary methods were used to assess predictor importance: impurity-based importance from the Random Forest model and permutation-based importance using ROC–AUC as the evaluation metric.

```
varImpPlot(rf_final, main="Random Forest Variable Importance (Gini Index)")
```

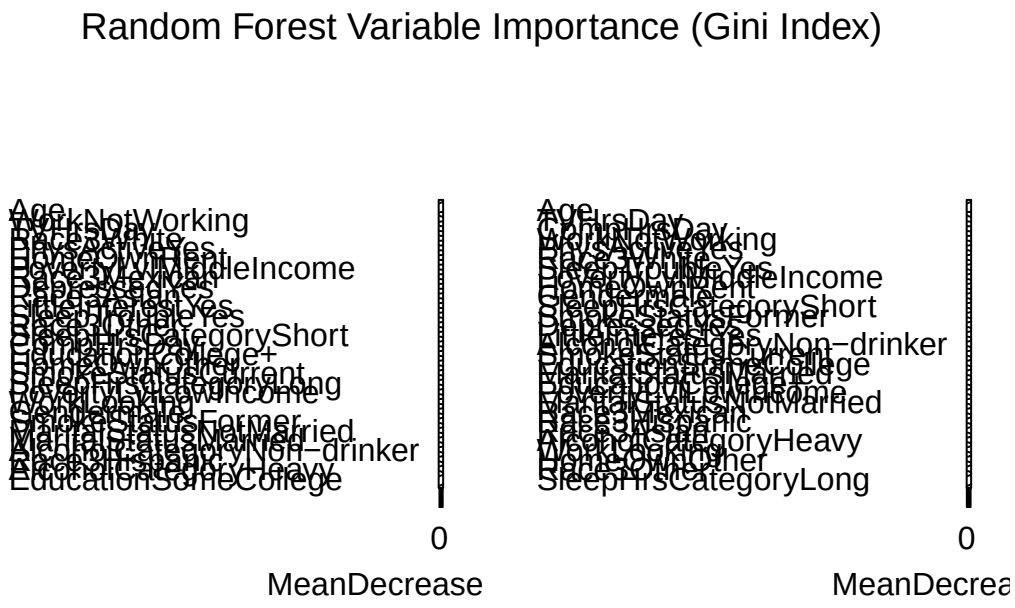


Figure X. Random Forest variable importance using Gini impurity reduction. Larger values indicate greater contribution to tree-based splits.

```
vip(vi, num_features = 15, geom = "col")
```

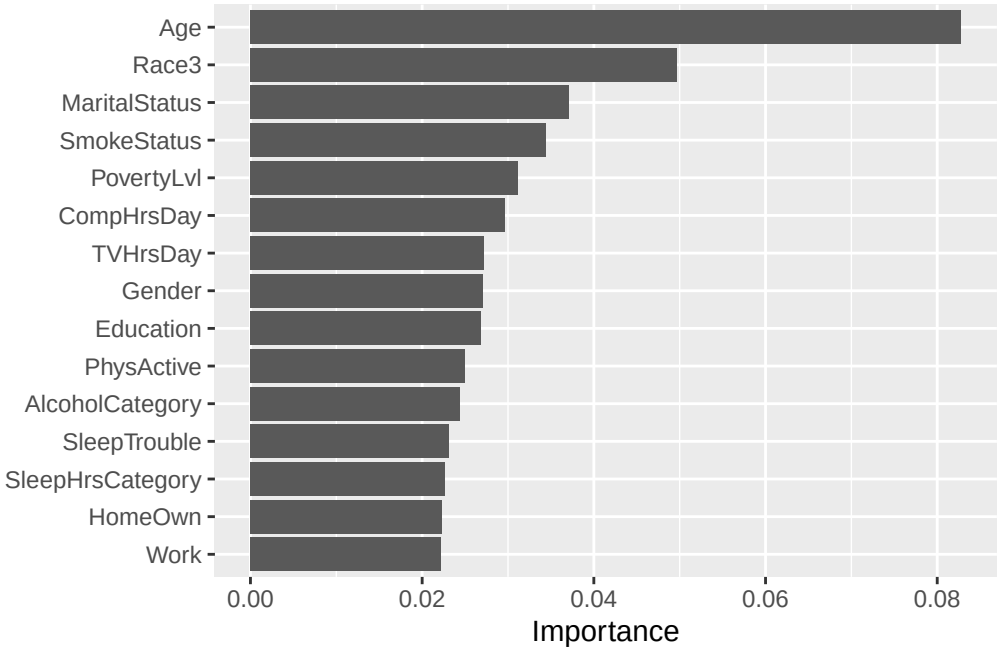


Figure X. Permutation-based feature importance ranked by AUC degradation. Bars represent performance loss after random permutation of each variable.

Clinical variables dominated both rankings. Body Mass Index, age, direct cholesterol, systolic blood pressure, diastolic blood pressure, and self-rated health consistently produced the largest reductions in model performance when permuted. This indicates that the classifier relied primarily on physiological and metabolic indicators when predicting diabetes status.

By contrast, socioeconomic characteristics (education, income, employment), behavioral factors (smoking, alcohol use, physical activity), and psychosocial measures (depression, loss of interest) contributed little to predictive performance when clinical variables were included.

This pattern closely aligns with the logistic regression findings. Both modeling strategies highlight the central role of metabolic health in diabetes risk among older adults. Rather than uncovering new predictors, the Random Forest model improved detection accuracy while reinforcing known clinical risk structures.

Comparison of Variable Importance

[Need revise if use Gini or both]

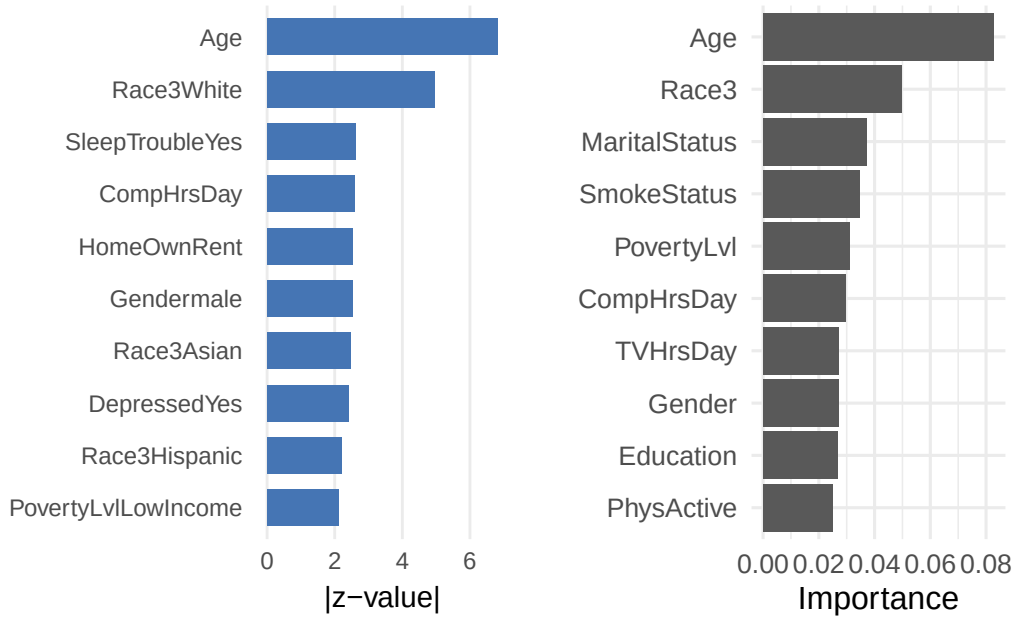


Figure 6: Comparison of Top 10 Variables Identified by Logistic Regression (Left) vs. Random Forest (Right)

Figure 4 compares the top 10 variables from logistic regression (ranked by |z-value|) with those from Random Forest (ranked by permutation-based importance). Strong concordance was observed between the two methods. Age dominated both rankings with the highest importance, followed by race (White vs. Black) in second position. Of the top 10 predictors identified by Random Forest, 5 overlapped with the logistic regression top 10, including age, race, poverty, gender, and computer usage hours. All 5 overlapping variables were statistically significant ($p < 0.05$) in the logistic regression model.

Notable differences emerged in mid-tier rankings. Sleep trouble and home ownership ranked 3rd and 5th in logistic regression (both $p < 0.01$) but did not appear in the Random Forest top 10. Conversely, marital status, smoking status, TV hours, and physical activity ranked prominently in Random Forest (3rd through 6th) despite showing weaker or non-significant associations in logistic regression. (This divergence may reflect Random Forest's ability to capture nonlinear relationships or complex interactions that logistic regression does not model explicitly.)

Socioeconomic indicators showed moderate importance in both models: poverty level appeared in both top 10 lists, while education ranked 7th in Random Forest but outside the top 10 in logistic regression. Home ownership (rent) ranked 5th in logistic regression but did not appear in the Random Forest top 10.

The substantial overlap in top-ranked variables strengthens confidence in these factors as robust diabetes predictors across both parametric and non-parametric modeling approaches.

Discussion

[need revise]

This study examined socioeconomic, behavioral, and psychosocial determinants of diabetes among U.S. adults aged 40 years and older using complementary analytical approaches. Both logistic regression and Random Forest modeling converged on age, race, poverty level, sleep-related factors, and sedentary behavior as robust diabetes predictors. Notable divergence between methods emerged for marital status, smoking, and physical activity, suggesting potential nonlinear relationships warranting further investigation.

Demographic and socioeconomic factors dominated both models. Age showed the strongest association (OR = 1.04 per year, $p < 0.001$), consistent with established links between aging and insulin resistance. Substantial racial/ethnic disparities persisted: Black participants exhibited higher diabetes odds than White (OR = 0.45, $p < 0.001$), Asian (OR = 0.48, $p = 0.013$), and Hispanic (OR = 0.55, $p = 0.028$) participants, corroborating prior NHANES analyses (Menke et al., 2015) and underscoring systemic inequities. Low income (OR = 1.49, $p = 0.034$) and renting versus owning a home (OR = 1.41, $p = 0.011$) independently predicted higher diabetes odds, consistent with pathways linking economic hardship to diabetes through food insecurity, chronic stress, and limited healthcare access (Hill-Briggs et al., 2021). Education showed no significant association in logistic regression despite moderate Random Forest importance, suggesting complex, nonlinear protective mechanisms.

Behavioral factors emerged as modifiable risk factors. Both short sleep duration (<7 hours; OR = 1.26, $p = 0.037$) and sleep trouble (OR = 1.36, $p = 0.009$) independently predicted diabetes, corroborating meta-analytic evidence that insufficient or poor-quality sleep disrupts glucose metabolism (Cappuccio et al., 2010; Shan et al., 2015). Computer usage hours demonstrated a dose-response relationship (OR = 1.13 per hour, $p = 0.010$), while TV viewing showed borderline significance (OR = 1.07, $p = 0.061$). Interestingly, Random Forest assigned TV hours higher importance, possibly reflecting nonlinear thresholds or interactions. Physical activity trended toward protection (OR = 0.80, $p = 0.057$) but did not reach significance, likely due to self-report limitations. Depression showed a significant positive association (OR = 1.43, $p = 0.016$), consistent with bidirectional links involving shared pathophysiology and behavioral pathways (Naicker et al., 2018).

Current smoking showed an inverse association (OR = 0.74, 95% CI: 0.55–0.99, $p = 0.046$), contradicting prospective evidence. This paradox likely reflects survival bias, older smokers with diabetes face elevated mortality, selectively removing them from the cross-sectional sample—and reverse causation, whereby individuals quit smoking after diagnosis. Current smokers with diabetes were younger than former smokers with diabetes (Supplementary Table

4), supporting mortality selection. Former smokers showed no protective association (OR = 1.07, p = 0.607). Our results should not be interpreted as evidence that smoking protects against diabetes.

Alcohol consumption showed no significant associations. Heavy drinking exhibited a non-significant inverse association (OR = 0.90, 95% CI: 0.53–1.46, p = 0.679), inconsistent with meta-analytic evidence of a U-shaped relationship (Knott et al., 2015). This null finding likely reflects limited power (small sample), substantial underreporting, and potential survival bias (Supplementary Table 5).

Comparison of logistic regression and Random Forest variable importance revealed both concordances and differences. Five variables (age, race, poverty, gender, computer usage hours) appeared in both top 10 lists, validating robustness. Marital status and smoking ranked highly in Random Forest despite weak logistic regression associations, suggesting threshold effects or interactions. Conversely, sleep trouble and home ownership ranked highly in logistic regression but not Random Forest, possibly reflecting predominantly linear effects.

Limitations

Several limitations warrant consideration. The cross-sectional design precludes causal inference and is vulnerable to reverse causation (smoking, physical activity, depression). KNN imputation relied on the missing-at-random assumption, potentially violated for alcohol and smoking. Self-reported measures are prone to recall and social desirability bias. Computer usage hours, TV watching hours, and race data were completely absent in 2009-2010, introducing potential temporal bias. While Random Forest can detect interactions, our analysis did not explicitly identify specific interaction terms.

Implications and Future Directions

Our findings support multifaceted diabetes prevention strategies targeting modifiable risk factors—particularly sleep health, sedentary behavior reduction, and socioeconomic interventions—alongside continued efforts to address structural racism and health inequities. Future research should employ prospective designs to establish temporality, utilize objective measures (actigraphy for sleep/activity, biomarkers for stress), and explicitly model interactions identified by Random Forest to elucidate underlying mechanisms. Machine learning methods may also inform personalized risk stratification, identifying subgroups for targeted interventions.

Conclusion

Integrating logistic regression and random forest approaches, we identified age, race, poverty, sleep disturbances, and sedentary behavior as robust diabetes risk factors among U.S. adults

aged 40 and older. Addressing modifiable risk factors—particularly sleep health and sedentary behavior—alongside upstream social determinants remains critical for diabetes prevention in this high-risk population.

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